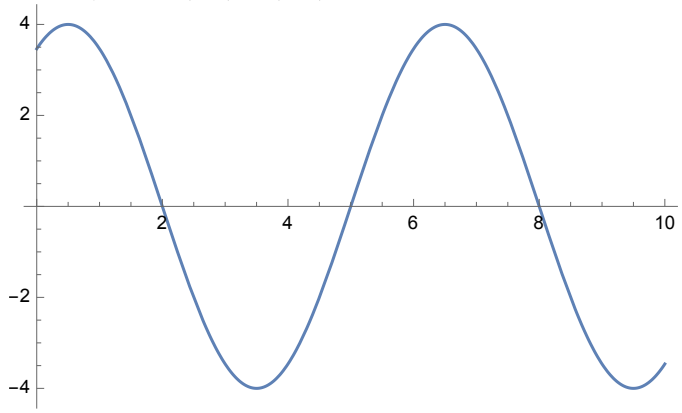
1. The graph below shows the displacement versus x coordinate for a wave moving to the right at 3 m/s. At (x,t) =(0,0) the wave has a displacement of $2\sqrt{3}$ m.



A. Find the frequency and wavelength of the wave.
B. Find the period of the wave
C. Find the angular frequency of the wave
D. Find the wave number of the wave
E. Find the initial phase of the wave
F. Find the amplitude of of the wave
G. Write the wave equation down

2. The general equation of a sine wave is $y = A\sin(kx + \phi_0)$. In geogebra, create sliders for k, ϕ_0 and A.

A. What happens when the wavenumber, k is increased?
B. When the wavenumber is increased, the wavelength _____
C. What happens when ϕ_0 is increased?
D. What happens when the amplitude is increased?

3. Suppose a wave has A = 1 m, λ = 3 m, and v = 6 m/s. In geogebra, type the following into the input bar to define the wave number, the angular frequency, and a function of both position and time:

```
k = 2 pi / 3
omega = k * 6
d(x, t) = sin(k x - omega t)
```

Then make a slider t that runs from 0 to 1 (with an increment of 0.01), plot `f(x) = d(x, t)`, and right-click the slider and choose **Animation On** to animate the wave over the first second.

A. How many full wavelengths fit in the window $0 \leqslant x \leqslant 12$ m?
B. How far does a crest travel in the first second? Does this agree with v = 6 m/s?
C. How many times does the wave repeat itself during that first second? What does that tell you about the period?